

ATPL EASA STUDY GUIDE | 040 - HUMAN PERFORMANCE

Chapter 10: Aeronautical Decision Making

Syllabus Source: CAE Oxford (pp. 319-354) | Pass Target: >= 90% AviationExam

1. AERONAUTICAL DECISION MAKING (ADM) & DECIDE MODEL

Aeronautical Decision Making (ADM) is the systematic, structured mental process used by flight crews to consistently determine the safest course of action in response to changing flight circumstances. The standardized FAA/EASA model is DECIDE:

DECIDE Step	Operational Action	Flight Deck Application
D - Detect	Detect the change	Notice that a change in aircraft status, weather, fuel, or operational environment has occurred.
E - Estimate	Estimate the significance	Assess the operational significance and urgency of the problem for flight safety.
C - Choose	Choose the desired outcome	Define the safe operational objective (e.g. divert to alternate aerodrome).
I - Identify	Identify options & actions	Determine actionable solutions to achieve the chosen outcome.
D - Do	Do the necessary action	Execute the selected solution decisively using CRM and checklist procedures.
E - Evaluate	Evaluate the outcome	Monitor and review whether the chosen action is solving the problem or requires revision.

2. SITUATIONAL AWARENESS (ENDSLEY'S 3-LEVEL MODEL)

Situational Awareness (SA) is having an accurate, dynamic mental model of the flight environment. Mica Endsley defined three hierarchical levels:

- **Level 1: Perception:** Perceiving the status, attributes, and dynamics of elements in the cockpit and environment (seeing altimeter reading, hearing ATC clearance, noticing warning lights). Loss of Level 1 accounts for ~76% of SA errors!
- **Level 2: Comprehension:** Integrating and synthesizing sensory data to understand what it means in relation to operational goals (e.g. realizing fuel burn is higher than planned due to headwinds).
- **Level 3: Projection:** The highest level: projecting the future status and trajectory of the aircraft (predicting that the flight will arrive below final reserve fuel if no diversion is initiated).

3. MASTER COGNITIVE BIASES & 'GET-THERE-ITIS'

- **Confirmation Bias:** The human tendency to seek, notice, and overvalue information that confirms pre-existing beliefs, while ignoring contradictory evidence (e.g. noticing one clearing sky patch while ignoring severe thunderstorm radar returns).
- **Plan Continuation Bias ('Get-There-Itis'):** Unconscious cognitive lock-in to complete a planned flight or landing despite deteriorating weather, marginal fuel, or unstable approach parameters.
- **Availability Heuristic:** Judging the probability of an event by how readily vivid past occurrences come to mind.

- Sunk Cost Fallacy: Justifying continuing a compromised flight because substantial effort, time, and fuel have already been expended.

4. THE FIVE HAZARDOUS ATTITUDES & ANTIDOTES (FAA / EASA)

Hazardous Attitude	Typical Pilot Thought Pattern	Mandatory Mental Antidote
1. Anti-Authority ('Don't tell me!')	Resents rules, regulations, and ATC instructions.	'Follow the rules; they are usually right.'
2. Impulsivity ('Do something quickly!')	Acts instantly without thinking or assessing options.	'Not so fast. Think first.'
3. Invulnerability ('It won't happen to me!')	Believes accidents only happen to others; takes risks.	'It could happen to me.'
4. Macho ('I can do it / Watch this!')	Takes unnecessary risks to impress others or prove superiority.	'Taking chances is foolish.'
5. Resignation ('What's the use?')	Feels helpless; abdicates control to luck or circumstances.	'I am not helpless. I can make a difference.'

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Algoritmo Paso a Paso (Guía para Dummies): SHELL Model & Error Analysis

REGLA MNEMOTÉCNICA: L-H-S-E-L Paso 1: Identificar siempre el Liveware Central (L) como el piloto/operador humano. Paso 2: Si el problema involucra mandos, pantallas o botones confusos -> Es un desajuste L-H (Hardware). Paso 3: Si involucra una checklist mal redactada o un manual confuso -> Es un desajuste L-S (Software). Paso 4: Si involucra turbulencia, calor o hipoxia en cabina -> Es un desajuste L-E (Environment). Paso 5: Si involucra mala coordinación entre Capitán y Primer Oficial o con ATC -> Es un desajuste L-L (Liveware).

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: In Mica Endsley's model of Situational Awareness, what represents Level 3 SA?

- [A]: Reading the altimeter accurately
- [B]: Comprehending the fuel flow rate
- [C]: Projecting the future status and position of the aircraft in time and space
- [D]: Executing a missed approach

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CORRECT: [C]. Level 1 is Perception; Level 2 is Comprehension; Level 3 is Projection of future status.

Q2: What psychological bias describes a pilot who continues an unstable visual approach in thick fog because

- [A]: Sensory adaptation
- [B]: Plan continuation bias ('get-there-itis')
- [C]: The Leans
- [D]: Hyperventilation syndrome

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Plan continuation bias is the unconscious tendency to adhere to an original flight plan despite evidence showing it is no longer safe.

Q3: What is the mental antidote for the hazardous attitude 'Anti-Authority' ('Don't tell me what to do!')?

- [A]: 'Not so fast. Think first.'
- [B]: 'Follow the rules. They are usually right.'
- [C]: 'It could happen to me.'
- [D]: 'Taking chances is foolish.'

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. The antidote for Anti-Authority is 'Follow the rules. They are usually right.'

Q4: Which step in the DECIDE aeronautical decision-making model involves evaluating whether the chosen course of action is the best?

- [A]: Detect
- [B]: Choose
- [C]: Do
- [D]: Evaluate

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CORRECT: [D]. Evaluate is the 6th and final step: closing the loop by assessing the actual consequences of the action.

Q5: What is 'Confirmation Bias' in flight deck operations?

- [A]: Always confirming ATC clearances on the radio
- [B]: Seeking only clues that support one's desired hypothesis while dismissing contradictory safety warnings
- [C]: Agreeing with the captain out of politeness
- [D]: Performing duplicate checklist verification

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Confirmation bias leads pilots to filter out warning signs that contradict their preconceived mental model.

Q6: The hazardous attitude 'Resignation' is characterized by which pilot mindset?

- [A]: Believing rules do not apply to them
- [B]: Believing that whatever happens is out of their control and leaving the outcome to fate
- [C]: Rushing into actions without evaluating consequences
- [D]: Showing off flying skills

[?] OFFICIAL RATIONALE & ANALYSIS - Official Rationale & Exam Strategy

CORRECT: [B]. Resignation occurs when a pilot abdicates responsibility, believing 'what's the use, it's out of my hands.' Antidote: 'I can make a difference.'